

Item 1

You are a lead investigator for the National Transportation Safety Board. A minibus traveling down a mountain road suffers brake failure. The driver reports swerving to avoid a stationary obstacle, causing the vehicle to slide sideways before colliding with a safety barrier. Your team must reconstruct the event to determine if safety protocols were followed. The minibus (mass 2500 kg) was traveling at an estimated 60 km/h when the driver spotted a large rock (mass 80 kg) that had fallen onto the road 50 meters ahead. The driver swerved, causing the bus to slide sideways. Post-accident analysis shows the bus slid for 25 meters on the asphalt ($\mu_k = 0.45$) before hitting the barrier.

Task A

Analyze the sliding phase of the incident.

- a) Calculate the friction force that acted to slow the bus during the slide.
- b) Using the Work-Energy Theorem, determine the bus's velocity just before it hit the barrier.
- c) The driver claims he was not speeding. Based on your calculations, was the initial 60 km/h speed estimate plausible? Justify your conclusion by working backward from the impact point.

Task B

A junior investigator suggests the bus might have actually sideswiped the rock, knocking it clear, instead of just swerving.

- a) State the condition under which momentum would be conserved in a collision between the bus and the rock.
- b) Assuming a perfectly elastic, glancing collision where the rock was initially at rest, derive a general equation for the final velocity of the rock. Explain how you would use trace evidence (e.g., the rock's final position) and this equation to test the hypothesis.
- c) Compare this to a perfectly inelastic collision. Which scenario would result in more severe damage to the bus? Explain using the concept of kinetic energy transfer.

Task C

The bus rolled slightly during the slide but did not tip over. Your team is tasked with recommending a safer vehicle design.

- a) Define the term centre of gravity. Describe a simple experiment to locate the centre of gravity of a scale model of the bus.
- b) Explain, using the concept of a couple, how a lower centre of gravity contributes to vehicle stability during a turn or slide.



Item 2

You are a thermal engineer hired by the "KATUBA" juice factory. Their current cooling system is inefficient, leading to product spoilage and high energy costs. The factory uses a vapor-compression cycle, but the core component—the evaporator—is underperforming. You must analyze and redesign this system. The evaporator is supposed to rapidly cool a batch of juice from 85°C to 5°C. The current system uses a refrigerant, but management wants to explore using pure water as a coolant in a new design, taking advantage of its high latent heat.

Task A

To calibrate your new sensors, you need a highly accurate thermometer.

a) Define a thermometric property and explain why the triple point of water is a superior fixed point for calibration compared to the ice point.

b) Describe the step-by-step process of establishing the centigrade scale using a constant-volume gas thermometer. A junior technician is getting inconsistent readings between two thermometers. Provide two plausible explanations for this discrepancy.

Task B

You are designing the new evaporator where the coolant (water) boils at low pressure.

a) Using the kinetic theory of matter, explain what is happening at the molecular level inside the evaporator as the liquid coolant boils.

b) The project manager questions why you need such a powerful heater to vaporize the coolant. Explain why the latent heat of vaporization is so much higher than the latent heat of fusion for the same substance.

c) Describe an experiment you would set up in the factory lab to accurately determine the specific latent heat of vaporization of the proposed water coolant using the method of mixtures. List the key measurements you would take.

Task C

You test a prototype cooling system that uses an electric heater to vaporize the coolant. In two separate trials, you collect the following data over 5-minute intervals:

Trial	Heater Voltage (V)	Heater Current (A)	Mass of Coolant Vaporized (g)
1	12.0	8.5	120
2	15.0	10.2	195



- a) Calculate the power input for each trial.
- b) From this data, calculate the average power required to vaporize one gram of the coolant per second. This value is critical for sizing the factory's main heater.
- c) The factory wants a system that can vaporize 5.0 grams of this coolant per second. Using your results, calculate the electrical power rating (in kW) the main heater must have.

"The journey of learning is not about reaching a destination, but about expanding the horizons of what you can achieve. You have overcome this challenge; carry this resilience forward into the next."

Malaba
Shafic

